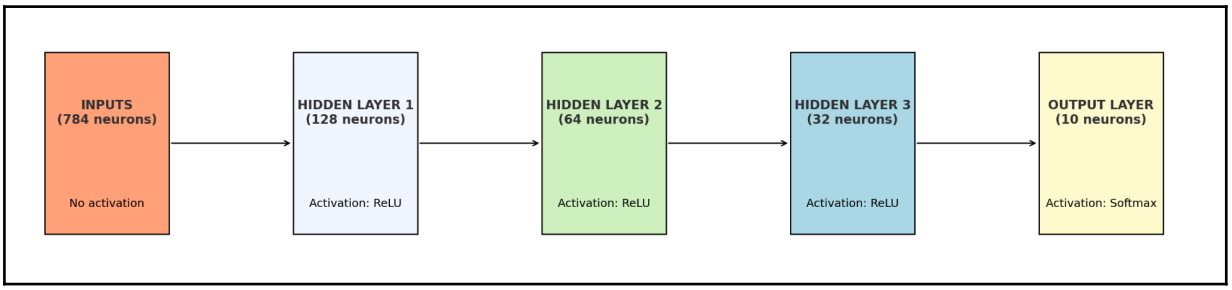
REPORT PROJECT 5

Modified National Institute of Standards and Technology database (MNIST) :

Task 1 : What is the structure of the ANN ?

The structure of the Artificial neural networks used in the algorithm is:

- **Input Layer**: with 784 neurons (28×28 pixels image, flattened to 784).
- **Hidden Layers**:
  - **First Hidden Layer**: with 128 neurons, activation function: ReLU.
  - **Second Hidden Layer**: with 64 neurons, activation function: ReLU.
  - **Third Hidden Layer**: with 32 neurons, activation function: ReLU.
- **Output Layer**: with 10 neurons (corresponding to digits 0 to 9) and the activation function is Softmax .



Remarks:

The activation function Relu doesn't change the positive values and transforms any negative value to 0.

$$f(x) = \begin{cases} o & for\ x \leq 0 \\ x & for\ x > 0 \end{cases}$$

The softmax activation function converts the values into probabilities between 0 and 1, and the sum of all probabilities is exactly 1.

$$f(z)_j = \frac{e^{z_j}}{\sum_{k=1}^{K} e^{z_k}}$$

Task 2 : What are the equations that are used to update the weights ?

The weights of the network are updated using Stochastic Gradient Descent (SGD) (optimization algorithm used to minimize a loss function, which represents how far the model's predictions are from the actual values). The equation used to update a weight is:

$$w_j := w_j + \Delta w_j \; with \; \Delta w_j = \eta \,.\, (target(i) - output(i)) \,.\, x_j^i$$

where :

η is the learning rate (set by default at 0.01),

$(target(i) - output(i))$ is the difference between the predicted value and the actual value.

and since x^i is a vector of length 784, x_j^i is the pixel value at position j.

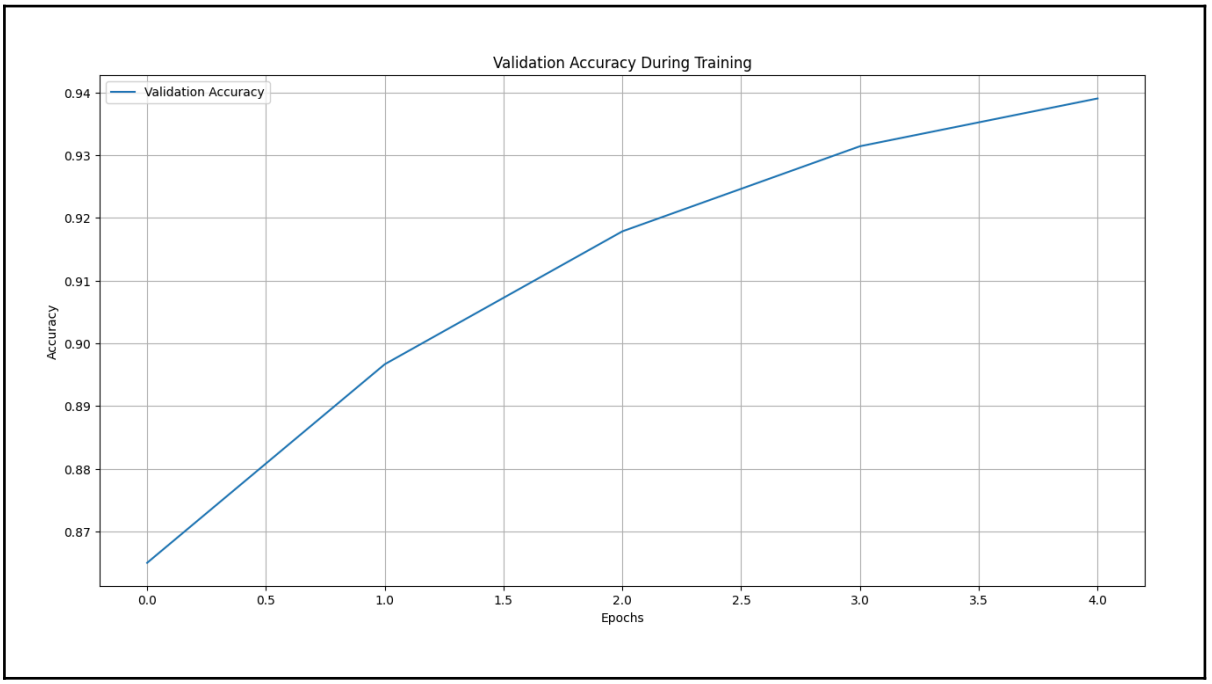
Source : What are gradient descent and stochastic gradient descent? by Sebastian Raschka

Task 3 : What is the percentage of correctness of the total test data set and the percentage of correctness of each of the classes in the test data set ?

```
test
263/263 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.9319 - loss: 0.2178
Overall test accuracy:  {93.41744780540466}
263/263 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step
Accuracy for class 0: 94.95%
Accuracy for class 1: 97.89%
Accuracy for class 2: 95.76%
Accuracy for class 3: 94.22%
Accuracy for class 4: 91.74%
Accuracy for class 5: 87.55%
Accuracy for class 6: 97.23%
Accuracy for class 7: 93.33%
Accuracy for class 8: 89.78%
Accuracy for class 9: 90.36%
```

We can see that the total correctness of the test date is approximately **93.41%**.

Task 4 : How does the accuracy in the validation set change during the training process ?



As the number of epochs increases, validation accuracy steadily improves, starting at approximately 86.5% and reaching 93% after 5 epochs. This means that the model is learning effectively from the training data and generalizing well to the validation set.